Question ID: 8f3a200e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3.

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3?

Answer

A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.

B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.

C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.

D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

Correct Answer: C

Rationale

Choice C is the best answer because it states a conclusion the researchers likely agree with, given the details in the text. The text explains that a biosignature gas is a gas that can be used as an indicator that a planet harbors some form of life and some astronomers have proposed that NH_3 could serve as a biosignature gas. The researchers evaluating this claim found that the atmosphere of rocky planets would be unlikely to reach "detectably high levels" of NH_3 without biological activity, which would support the proposal of NH_3 serving as a biosignature gas. However, the text also states that mini-Neptune planets can produce NH_3 in the absence of biological activity. Thus, the text is structured to lead to the conclusion that detectable levels of NH_3 in the atmospheres of rocky planets could constitute a biosignature, but that is not the case for detectable levels of the gas in the atmospheres of mini-Neptune planets.

Choice A is incorrect because the text indicates that biological activity likely accounts for detectable levels of NH_3 in the atmospheres of rocky planets but mini-Neptune planets can have detectable levels of NH_3 in their atmospheres in the absence of biological activity. Therefore, both rocky planets and mini-Neptune planets can have detectable levels of atmospheric NH_3. Choice B is incorrect because the text states that for NH_3 to reach detectable levels in the atmospheres of rocky planets likely means they harbor biological activity, meaning that rocky planets with detectable NH_3 usually harbor biological activity. However, that does not entail that every rocky planet with biological activity will have detectable levels of NH_3 in their atmospheres. Choice D is incorrect because the text claims only that some astronomers have proposed using NH_3 as a biosignature gas without mentioning a minimum concentration of atmospheric NH_3 that must be met for it to function as a biosignature gas.